ILLINOIS TECH | College of Computing

# ITMM 485 / 585

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Legal and Ethical Issues in Information Technology

ILLINOIS INSTITUTE OF TECHNOLOGY 1890

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# Contract Law:
# SLAs and Outsourcing
# PI: Outsourcing, SLA definition

# Announcements

- Internet 02 section- clarification
- Final Exam

# Learning Objectives

Upon completion of this lesson the students should be able to:

- Define a Service Level Agreement (SLA)
- Identify and discuss the goals of a Service Level Agreement
- Explain the RFQ/RFP process and how it relates to Service Level Agreements
- Describe and discuss the expected contents of a Service Level Agreement
- Identify and explain key questions that should be addressed when drafting an SLA

# Agenda

- Outsourcing Overview
- Contract Law & IT
- SLA Fundamentals
- RFQ/RFP Process
- SLA Structure & Metrics
- Key Legal Questions

# Contract Law in IT

- IT operations and services governed by contracts
- SLAs serve as specialized contracts for services


THE BIGGEST AWS OUTAGE IN HISTORY:
THE DECEMBER 7, 2021 US-EAST-1 MELTDOWN
AND LESSONS LEARNED
aws

# AWS Outage (2021)

- Major cloud outage impacting multiple services
- Affected companies: Netflix, Disney+, Robinhood, etc.
- Duration: Several hours
- Root cause: Failure in AWS internal network services

## The Most Common IT Support Metrics

### Cost

- Cost per Ticket/Contact
- Cost per Minute of Handle Time
- First Level Resolution Rate

### Quality

- Customer Satisfaction
- First Contact Resolution Rate
- Call Quality

### Productivity

- Inbound Contacts per Analyst per Month
- Analyst Utilization
- Analysts as a % of Total Headcount

### Analyst

- Annual Analyst Turnover
- Daily Analyst Absenteeism
- Schedule Adherence
- New Analyst Training Hours
- Annual Analyst Training Hours
- Analyst Tenure
- Analyst Job Satisfaction

### Service Level

- Average Speed of Answer (ASA)
- % of Calls Answered in 30 seconds
- Call Abandonment Rate

### Call Handling

- Inbound Contact Handle Time
- User Self-Service Completion Rate

# What is Outsourcing?

Delegation of IT functions to external providers

# Why Outsource? The Reasons

- Cost and Quality concerns
- Breakdown in IT performance
- Intense vendor pressures
- Simplify management
- Financial factors
- Corporate Culture
- Eliminate internal irritant
- Other factors

# Cost & Quality

- Cost control of benefits
- Use of low-cost labor pools in India, Ireland, Poland, etc.
- Higher standards for outsourced IT staff
- Economies of scale through bulk purchasing/licensing
- Better management of hardware capacity
- More aggressive management of service & response time
- Leaner management structure
- Better IT staff skill sets

# Breakdown in IT Performance

- Dysfunctional IT departments
- Corporate neglect of IT
- Inadequate staffing
- Inability to manage technical field outside line of business
- “Cost savings”
- IT staff “do not understand line of business”
- Outdated skill sets in staffing
- Others?

# Vendor Pressures

- Large vendors offer turnkey solutions
- Lower costs with higher efficiency

# Management Simplification

- Allows focus on core business
- Eliminates non-core functions

# Financial Motives

- Liquidate tangible and intangible IT assets
- Convert fixed cost expense to variable cost
- Allows flex based on required real capacity rather than peak capacity
- “Pay-as-you-go” make IT expenses “hard-dollar” instead of “soft-dollar” expenses
- Also referenced as OpEx versus CapEx(Operational Expenses versus Capital Expenses)
- In most firms, accounting folks seem to think OpEx trumps CapEx

# Other Outsourcing Factors

- Corporate culture and fit
  - "Geeks" may not be good fit in organizational culture
- Eliminate internal irritant
  - Poor perception of IT performance
  - Lack of career opportunity/upward mobility for IT staff
- Others?

# When to Outsource

- Based on strategic grid
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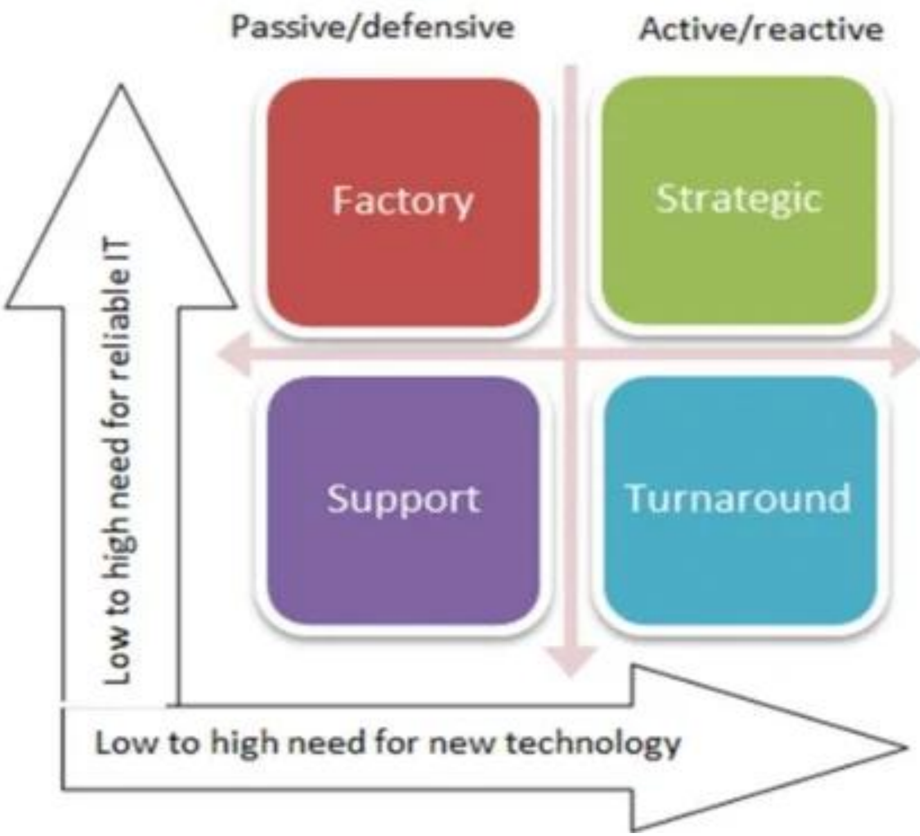
# Strategic Grid Positioning




When a company uses IT to obtain both operational and competitive benefits through information and its IT, it is located in the **Strategic** quadrant of the Grid.

When IT contributes to operational efficiency and reliable service delivery, the organization is in the **Factory** quadrant of the grid

The **Support** quadrant is where the company is the least dependent on IT/ICT, i.e., where few operational benefits are being achieved from the organization's IT

The **Turnaround** quadrant is where investment is new systems promise major process and service efficiencies, cost reduction and enhanced or new markets.

The strategic grid framework (based on Nolan & McFarlan, 2005).

# Structuring Outsourcing

- Contract flexibility
- Standards & control
- Determining areas to outsource
- Easy separation
- Specialized competencies
- Centrality of activities
- Cost savings
- Supplier stability & quality
- Management fit
- Conversion problems

# Risks in Outsourcing (1)

- Treating IT as a commodity
- Incomplete contracting
- Lack of vendor attention once contract is awarded
- Power goes to the vendor
- Inexperienced staff
- Outsourcing for short-term gain rather than business advantage
- Hidden costs

# Risks in Outsourcing (2)

- Loss of innovation
- Managing multiple vendors
- Cultural incompatibility

# Role of CIO in Outsourcing

- Contract management
- Architecture planning
- Technology and vendor oversight
- Continuous learning

# Managing Outsourcing

- Performance measurement
- Determination of and agreement on metrics, which are spelled out in Service Level Agreements
- Mix & coordination of tasks
- Customer-vendor relationship
  - Executive/management levels
  - Relationship managers
  - Coordinators/integrators

# Introduction to SLAs

- Collaborative written agreement between a customer and an IT service provider
- Defines key targets and responsibilities between both sides
- Provides framework for operation and common understanding for both parties
- Most effective when both sides actively collaborate on what should be included

# SLA Definitions

- TM Forum defines SLAs as expectations among two or more parties regarding service quality, priorities, and responsibilities
- The Cloud Standards Customer Council defines cloud SLAs as written expectations for service between cloud consumers and providers
- ITIL v3 defines SLA as terms between a provider and a customer that describe a service, document targets and specify responsibilities

# SLA Goals

- Allow IT to understand customer service requirements
- Control customer expectations for levels of service to be delivered
- Allow clear understanding of priorities when service problems arise

# What are SLAs

- SLAs are business agreements
- A binding contract
- Written by the provider
- Need to meet customer expectations which may be specified by
  - Request for Quotation (RFQ –simple) or
  - Request for Proposal (RFP –complex)

# RFQ vs RFP

- Request for Quotation(RFQ) issued when scope of work or services is standard and likely to be provided as predesigned/preconfigured product
- Request for Proposal(RFP) issued when scope of work or services is specifically defined and requires custom input from vendors/providers

# RFPs

- Tool to collect information from vendors
- Vendors need to provide detailed info about each specific application/product in which interest is indicated
- Covers other areas: training, support, maintenance, contracting and references
- Gives vendor/provider a deadline
- Includes a contact name and phone number for follow-up questions

# RFP ?s

- Does the vendor have installations of similar size and configuration?
  - How many sites and where are they located?
- What types of updates to the product are currently in development? Is there a scheduled release date?
- What types of updates to the product are being planned? What is the largest number of users on one system?
  - What is the cost of adding items [a work station, another switch, etc]?
  - What is the cost of adding a customized feature?

# RFP ?s

- Is the vendor financially sound?
- What is the vendor's track record?
- Vendor's philosophy on problem solving, service, commitment to customer satisfaction
- Who "owns" the data?
  - Who has access to it?
  - Under what conditions?

# RFP ?s

- Does the vendor provide on-site training?
- Does the vendor provide acceptable training/reference manuals?
- How is documentation (user’s manual/information) provided? Is it written clearly and is it easy to understand?
- Is it available on-line?
- Is there a trouble-shooting section?
- How frequently is the manual updated?
- Do users routinely receive updated copies?

# RFP ?s

- What are the maintenance arrangements and costs? Annual maintenance agreement/cost?
- Fee for support service on an hourly basis?

# Process

- Get Proposals
- Validate responses
- Check  w. Reference customers
- Get “final and best offer” from top 2 or 3 vendors
- Make selection based on criteria (AHP)

# Keep SLA simple, measurable & realistic

- Use clear, concise wording; avoid ambiguity
- Avoid legal and technical jargon as well as unnecessary technical terminology
- Provide a glossary of terms as necessary
- Have someone independent from the process review the SLA
- Have legal counsel review it as well

# SLA Contents

- General Overview
- Description of Services
- Service Performance Level
- Service Provider/Customer Responsibilities
- Problem Management & Disaster Recovery Process
- Periodic Review Process
- Termination of Agreement Process
- Costs
- Signatures

# SLA Contents: Description of Services

- What systems are supported?
- What services are included?
- What services are NOT included?
- How will service be delivered?
- What are the hours of operation (regular business hours and after-hours support)?
- When will regularly scheduled maintenance be performed?

# SLA Contents: Service Performance Level

- Proper metrics are key to SLA success
- Response Time example: 95% of users will experience a response time of two seconds or less during regular working hours of 7:30 to 5:00

# SLA Contents: Service Performance Level

- Throughput example: Defines rate that data is delivered, i.e. a file transfer/ download of at least x mb will be transferred in y minutes
- Utilization defines maximum usage where system will perform within guaranteed response times and throughput; i.e. may specify maximum number of simultaneous users

# SLA Contents: Service Performance Level

## Customer Support

- Includes typical help desk problem reporting and problem resolution guarantees based on severity level
- Severity level and response and resolution times are assigned according to impact on customers
- Acceptable response time / resolution time negotiated between the IT Service Provider and the Customer

# SLA Contents: Service Performance Level

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# SLA Contents: Service Performance Level

## Availability

- Includes system availability guarantees over a period of time, i.e. application will be available 98%, 7 days a week, 19 hours per day
- Often expressed in “nines”

# Availability

| Availability | Minutes of downtime per year (24*365 operations) |
|---|---|
| 90.000% | 52,560.00 |
| 99.000% | 5,256.00 |
| 99.900% | 525.60 |
| 99.990% | 52.56 |
| 99.999% | 5.26 |

# SLA Contents: Service Performance Level

- Response Times
- Availability Windows
- Equipment Service and Repair Times
- Technical Support Response and Level
- Report or Other Media Delivery
- Security Access
- Data Retention / Backup Requirements

# Provider Responsibilities - examples

- Meet response times associated with service related incidents
- Generate service level reports for customer
- Train required staff on appropriate service support tools
- Notify customer of all scheduled maintenance
- Develop and maintain system documentation (could also be customer responsibility)
- ?
- Manage user accounts

# Customer Responsibilities - examples

- Adhere to policies, processes and procedures
- Report problems using procedures as per SLA
- Schedule in advance all service related requests & other special services with provider
- Develop and maintain system documentation (could also be service provider responsibility)
- Make customer representative available when resolving service related incident or request
- Communicate when system testing and/or maintenance could interfere with standard business functions

# SLAs / Penalties / Credits

## Will the SLA include service level credits?

- "service level credit" is a credit the supplier grants to the customer after a service level failure
- Supplier may be required to write a check to the customer or the customer may simply have the right to apply the credit to future service
- Reduces the effective price of the services and the supplier's profit margin
- SLAs can be terminated for cause upon a material breach (breach of contract, not data!)
  - A sufficiently severe service level failure would be a material breach

# AWS Outage (2017 – S3 Service Disruption)

## What happened:

- Amazon Web Services S3 outage (US-East-1)
- Took down thousands of websites (Slack, Trello, Quora)
- Lasted ~4 hours
- SLA Violation: AWS S3 SLA promised 99.9% uptime

# AWS Outage (2017 – S3 Service Disruption)

## What happened:

- Penalty: Customers received service credits (10–25%)
- Credits applied to future bills only
- Business losses (millions) ≠ SLA compensation (small credits)

# Azure Outage (2020 – Authentication Failure)

## What happened:

- Microsoft Azure Active Directory outage
- Users couldn’t log into Office 365, Teams, and other services

# Azure Outage (2020 – Authentication Failure)

**What happened:**

- SLA Violation Identity services unavailable → breach of uptime guarantees
- Penalty: SLA credits issued (typically 10–25% of monthly cost)
- Critical services failed, but financial penalties were minimal

# Common SLA Penalty Structure

- **Typical SLA Credits:**
  - 99.9% uptime breached → 10% credit
  - 99% uptime → 25% credit
  - Severe failure → 50% credit (rare)
  - Credits apply only to future invoices
- **No compensation for:**
  - Lost revenue
  - Reputational damage
  - Customer churn

# ?